

PROJECT SUMMARY

Overview:

Artificial Intelligence (AI) is transforming scientific discovery across domains, including biology, medicine, climate science, materials science, and physics. Despite recent advances, the integration of AI into large-scale scientific data analysis and visualization remains in its early stages. Major challenges persist in bridging scientific data, domain experts, and AI models in ways that support scalable, interpretable, and interactive scientific discovery. In particular, modern scientific workflows generate increasingly massive, high-dimensional, multimodal, and time-varying data that exceed the capabilities of existing analysis and visualization paradigms. These challenges highlight the need for next-generation AI-driven scientific systems capable of understanding, reasoning about, and interactively exploring complex scientific data at scale. In this context, the proposed research establishes foundational methodologies for AI-driven scientific data analysis and visualization systems that integrate generative AI, agentic reasoning, and scalable data representations. The vision is to create next-generation intelligent scientific discovery frameworks that combine advanced neural rendering, multimodal large models, and AI-friendly data representations to enable expressive, explainable, and scalable scientific analysis over large-scale and in situ scientific datasets. The proposed research pioneers algorithmic and systems innovations targeting the convergence of artificial intelligence, scientific visualization, and high-performance computing for accelerating scientific discovery.

Intellectual Merit:

This project is centered on the analytical, algorithmic, and systems foundations of AI-driven scientific data understanding and discovery automation. The overarching goal is to establish a cohesive framework that enables scalable scientific data analysis, intelligent visualization, and autonomous exploratory reasoning through the integration of generative AI, multimodal learning, and high-performance computing infrastructures. The proposed research addresses fundamental challenges in scientific data representation, interactive exploration, and automated discovery workflows under extreme-scale computational environments.

To this end, the research objectives are organized into three intertwined thrusts.

(T1) Transformative integration of data enabling science through semantically rich embeddings, linked scientific artifacts, and federated multimodal data management frameworks for scalable scientific knowledge discovery and reuse.

(T2) Intelligent data exploration assistants for science based on AI-enhanced visualization, implicit neural representations, and multimodal large-model reasoning for scalable, interactive, and context-aware scientific data understanding.

(T3) Automatic data analysis frameworks for accelerating scientific discovery through agentic AI systems, cloud-based high-performance computing infrastructures, and automated experimental analysis and visualization pipelines across diverse scientific domains.

Broader Impacts:

Given the ubiquity of scientific data and the growing importance of AI-assisted discovery, the utility of the proposed research extends well beyond scientific visualization to broader fields, including artificial intelligence, data science, high-performance computing, and computational science. By enabling scalable and intelligent analysis of large-scale scientific data, this project offers the potential to significantly accelerate discovery in critical domains such as biotechnology, medicine, climate science, chemistry, and materials science. The proposed research will also contribute broadly to education and workforce development through interdisciplinary student training at the intersection of AI and science, graduate mentoring in AI-driven scientific computing, undergraduate research experiences in visualization and machine learning systems, and outreach activities designed to broaden participation in computational science and artificial intelligence. The resulting open frameworks, methodologies, and infrastructures will promote more accessible, efficient, and inclusive scientific discovery for the broader research community.